Topic: Statistical Representation of Random Vectors

Definition: For random variables $X_1, X_2, \ldots, X_N$, we represent them as a random vector $\underline{X} = [X_1, X_2, \ldots, X_N]^T$.

- **Mean Vector:** Represents the average value of the random vector:

$$\underline{\mu}_X = E[\underline{X}] = [E[X_1], E[X_2], \ldots, E[X_N]]^T$$

- **Correlation Matrix:** A symmetric matrix $R_{XX} = E[\underline{X}\underline{X}^T]$ where elements are defined as:

$$R_{XX}(i,j) = E[X_i X_j] \quad \text{for } i,j \leq N$$

- **Covariance Matrix:** Defined as the expected outer product of the deviations from the mean:

$$C_{XX} = E[(\underline{X} - \underline{\mu}_X)(\underline{X} - \underline{\mu}_X)^T]$$

  It acts as a symmetric "mirror" mapping dependencies where $Cov(X_i, X_j) = Cov(X_j, X_i)$.

Topic: Linear Transformation of Random Vectors

Definition: A linear transformation is defined by the model:

$$\underline{Y} = \underline{A}\underline{X} + \underline{b}$$

Where $\underline{A}$ is the transformation matrix (scaling/rotation), $\underline{b}$ is a constant vector (shift/translation), and $\underline{X}$ is the input random vector.

Key Results

- **Mean Transformation:** $\underline{\mu}_Y = \underline{A}\underline{\mu}_X + \underline{b}$
- **Covariance Transformation:** $C_{YY} = \underline{A} C_{XX} \underline{A}^T$
- **Correlation Transformation:** $R_{YY} = \underline{A} R_{XX} \underline{A}^T + \underline{A}\underline{\mu}_X \underline{b}^T + \underline{b}\underline{\mu}_X^T \underline{A}^T + \underline{b}\underline{b}^T$

Derivations

Proof: Covariance Matrix Transformation

1. Observe that the deviation from the mean is independent of the shift $\underline{b}$:

$$\underline{Y} - \underline{\mu}_Y = (\underline{A}\underline{X} + \underline{b}) - (\underline{A}\underline{\mu}_X + \underline{b}) = \underline{A}(\underline{X} - \underline{\mu}_X)$$

2. Substitute into the covariance definition:

$$C_{YY} = E[(\underline{Y} - \underline{\mu}_Y)(\underline{Y} - \underline{\mu}_Y)^T] = E[\underline{A}(\underline{X} - \underline{\mu}_X)(\underline{X} - \underline{\mu}_X)^T \underline{A}^T]$$

3. Factor out the constant matrix $\underline{A}$:

$$C_{YY} = \underline{A} E[(\underline{X} - \underline{\mu}_X)(\underline{X} - \underline{\mu}_X)^T] \underline{A}^T = \underline{A} C_{XX} \underline{A}^T$$

Conclusion: The shift $\underline{b}$ does not affect the covariance (spread).

Topic: Gaussian Distributions: 1D and Multivariate

- **Univariate Gaussian (1D):**

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- **Multivariate Gaussian PDF (N-dimensional):**

$$f_{\underline{X}}(\underline{x}) = \frac{1}{\sqrt{(2\pi)^N \det(C_{XX})}} \exp\left(-\frac{1}{2}(\underline{x}-\underline{\mu}_X)^T C_{XX}^{-1}(\underline{x}-\underline{\mu}_X)\right)$$

Example: Uncorrelated Gaussian Random Variables
If variables are mutually uncorrelated, $Cov(X_i, X_j) = 0$ for $i \neq j$. C_{XX} becomes a diagonal matrix, and the joint PDF factorizes:

$$f_{\underline{X}}(\underline{x}) = \prod_{n=1}^{N} \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp\left(-\frac{(x_n-\mu_n)^2}{2\sigma_n^2}\right)$$

Case Study: Smart City Traffic & Weather System

Context: Predicting traffic congestion in Ahmedabad using an N-dimensional random vector $\underline{X}$ (flow, speed, rainfall, temp, AQI). **Model:** $\underline{X} \sim \mathcal{N}(\underline{\mu}, \Sigma)$.

Linear Transformation Example:
Define a **Congestion Index** $Y = \underline{a}^T \underline{X} = 0.6X_1 - 0.4X_2 + 0.8X_3$. Since $\underline{X}$ is Gaussian and Y is a linear transformation:

- Y is also Gaussian: $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$
- $\mu_Y = \underline{a}^T \underline{\mu}$
- $\sigma_Y^2 = \underline{a}^T \Sigma \underline{a}$